



CS & IT ENGINEERING



Algorithms

Dynamic Programming

Lecture No.- 08



By- Aditya sir

Recap of Previous Lecture



Topic

Topic



Multi-Stage Graph

Topics to be Covered



Topic

Topic

Topic

TSP

PYQs



About Aditya Jain sir

1. Appeared for GATE during BTech and secured AIR 60 in GATE in very first attempt - City topper
2. Represented college as the first Google DSC Ambassador.
3. The only student from the batch to secure an internship at Amazon. (9+ CGPA)
4. Had offer from IIT Bombay and IISc Bangalore to join the Masters program
5. Joined IIT Bombay for my 2 year Masters program, specialization in Data Science
6. Published multiple research papers in well known conferences along with the team
7. Received the prestigious excellence in Research award from IIT Bombay for my Masters thesis
8. Completed my Masters with an overall GPA of 9.36/10
9. Joined Dream11 as a Data Scientist
10. Have mentored 12,000+ students & working professions in field of Data Science and Analytics
11. Have been mentoring & teaching GATE aspirants to secure a great rank in limited time
12. Have got around 27.5K followers on LinkedIn where I share my insights and guide students and professionals.



Telegram Link for Aditya Jain sir: https://t.me/AdityaSir_PW

TTTe



Topic : Dynamic Programming :(DP)

PYQ

A = "qpqrr" $\rightarrow$ 5

B = "pqprrqp" $\rightarrow$ 7

x = length of LCS

y = number of LCS

~~x = 10y?~~

$$x + 10y = ?$$

$$= 4 + 10 \times 3 = \underline{34}$$

HW $\rightarrow$ DP

$$\underline{x = 4}$$

LCS

- 1) qprrr
- 2) pqprrr
- 3) qprrr

$$\underline{y = 3}$$

27.8%

1 ✓	↓	5 ✗
2 ✓	↓	4 ✓
3 ✓	↓	

$$x=4$$



Topic : Dynamic Programming :(DP)

LCS: 1. ~~qprr~~

2. qpqr

3. pqrr

4. qprr

$$x = \text{len (LCS)} = 4$$

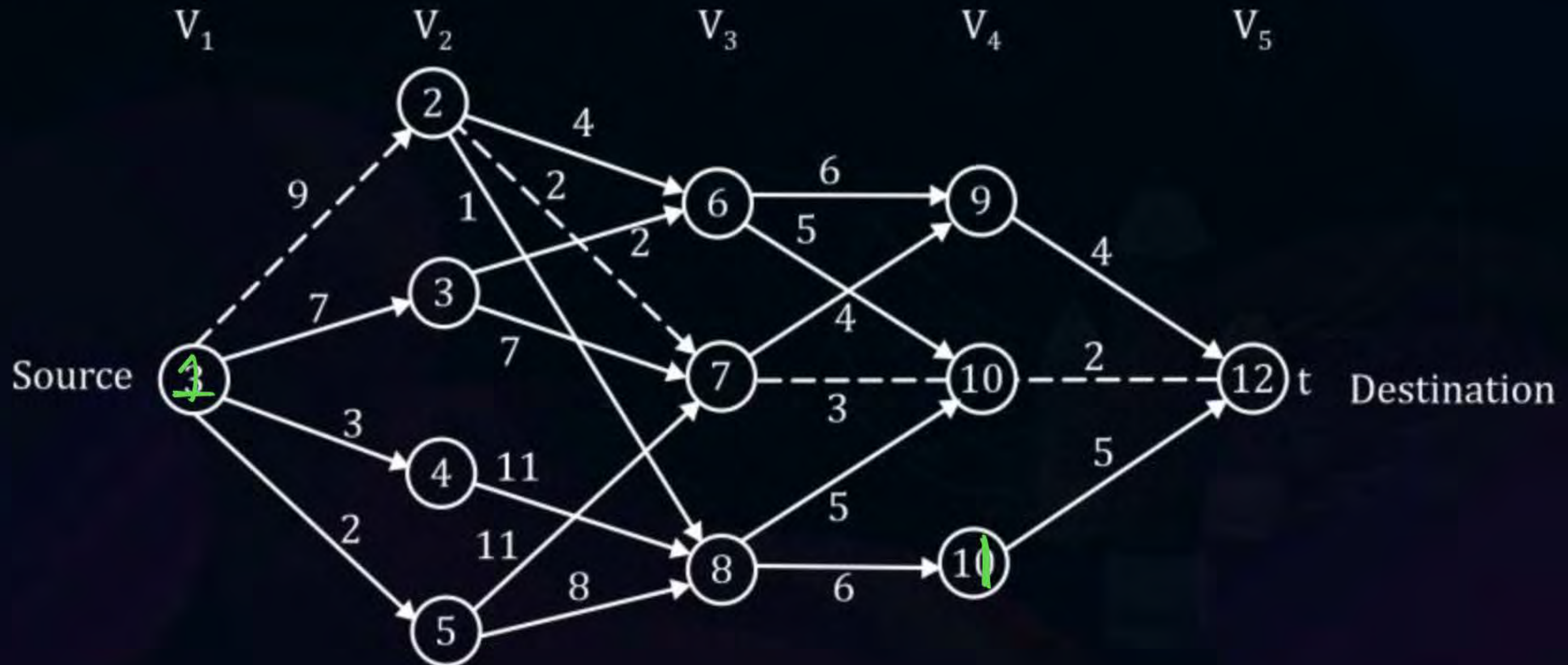
$$y = \text{number of LCS} = 3$$

$$\begin{aligned} x + 10y &= 4 + 10 \times 3 \\ &= 4 + 30 \\ &= 34 \end{aligned}$$



Topic : Dynamic Programming :(DP)

Multi-stage graph





Topic : Dynamic Programming :(DP)

Multi-stage graph

$$\text{cost}(i, j) = \min \{c[j, k] + \text{cost}(i + 1, k) \}$$

$$= K \in V_{i+1}$$

$$\langle j, k \rangle \in E$$

$$l \rightarrow \text{stage:} \quad \text{cost}(l-1, j) = C(j, t)$$

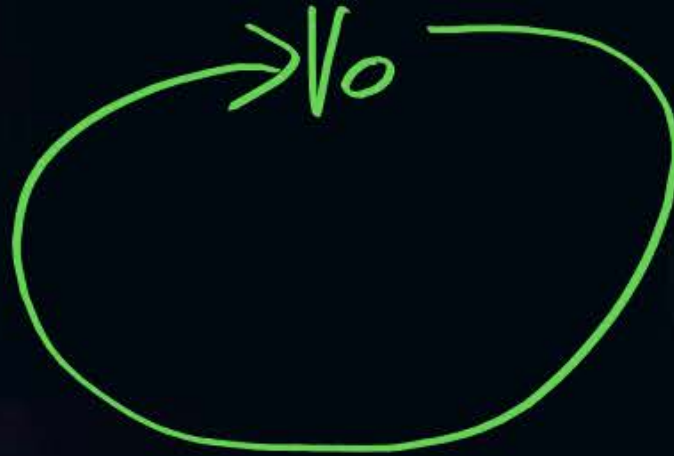
$$t \rightarrow \text{destination} \quad D(l, j) = K \rightarrow \text{that } k \text{ which minimize cost}(i, j)$$



Topic : Dynamic Programming :(DP)

8. Travelling Salesman Problem (TSP)

The tour of TSP should start from the home city (V_0) and visit remaining $(n-1)$ cities exactly once and come back to the home city (V_0), such that the cost of tour is minimize.





Topic : Dynamic Programming :(DP)

Home city = Starting vertex (V_0)

Tour: $V_0 \rightarrow \underline{n-1}$ (vertices) $\rightarrow V_0$

TSP $\rightarrow$ Optimization problem

Objective function $\rightarrow$ minimize the tour cost.



Topic : Dynamic Programming :(DP)

Greedy Method

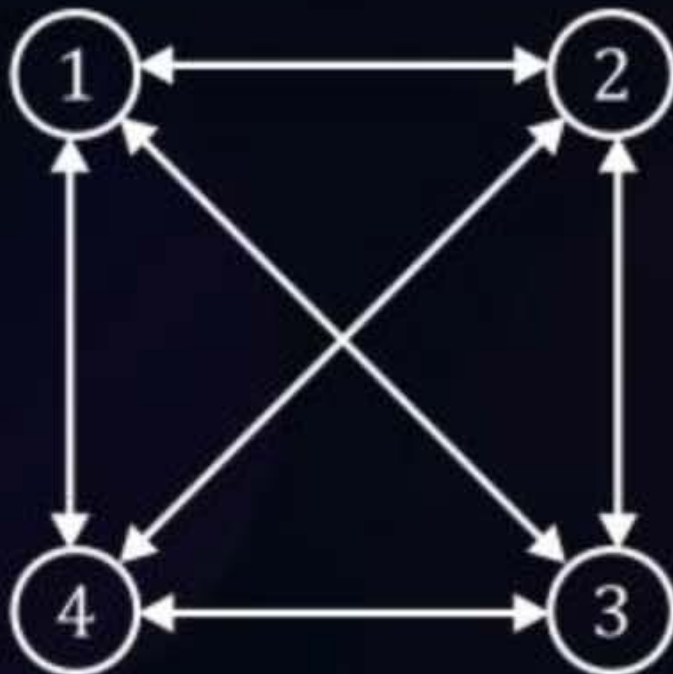
$$1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1$$

$$= 10 + 9 + 12 + 18$$

$$= 19 + 20$$

$$= 39 \checkmark$$

Source $\rightarrow$



Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0



Topic : Dynamic Programming :(DP)

The actual optimal/minimum tour cost is-

$$1 \rightarrow 2 \rightarrow 4 \rightarrow 3 \rightarrow 1$$

$$10 + 10 + 9 + 6$$

$$= 20 + 15$$

$$= 35$$

Optimal solution



Imp. Note:-

If we solve TSP by greedy, we get min tour cost as 39 $[1 \rightarrow 2 \rightarrow 3 \rightarrow 4 \rightarrow 1]$

Which is not optimal solution.

Hence, greedy fails for TSP.



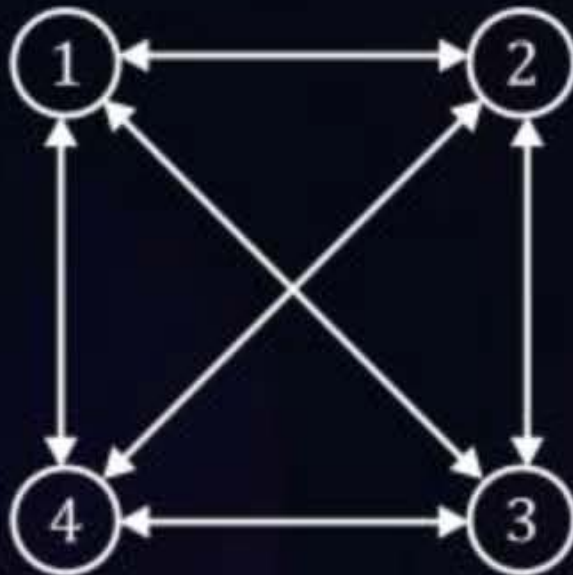
Topic : Dynamic Programming :(DP)

DP based solution:

Let $g(i, s)$ represent the cost of the ~~tour~~ of TSP from vertex ' i ' visiting ~~are~~ ^{all} the vertices in the set ' s ' exactly once and terminating the ~~tour~~ at V_0 (source)

Let, $V_0 = 1$ prev.. E.g.

E.g.





Topic : Dynamic Programming :(DP)

$$V_0 = 1$$

$$g(\underbrace{1, \{2, 3, 4\}}_S) = \min_{K \in \{S\} \text{ and } \langle 1, k \rangle \in E} \{ \underbrace{C(1, K) + g(K, S - \{k\})}_{\text{subproblem}} \}$$

Original Problem $\rightarrow g(1, s)$

$[1 \rightarrow \underline{k} \dots 1] \rightarrow$ Sub-problem

$$C(1, k) + g(K, S - \{K\})$$



Topic : Dynamic Programming :(DP)

Generalized from $\rightarrow$ V. imp

$$g(i, s) = \min [c(i, k) + g(k, \underline{s - \{k\}})] \dots\dots(1)$$

$$K \in S$$

$$\langle i, k \rangle \in E$$

$$g(i, \phi) = c(i, v_0)$$

Starting vertex.

To make sure back to source v_0

$\rightarrow$ Terminating/Base Condition.

null

$$i \rightarrow v_0$$

$$J(i, S) = \text{Value of } k \text{ that minimize equation (1)}$$



Topic : Dynamic Programming :(DP)

E.g.



Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Bottom - up

Optimal problem

$$g(1\{2,3,4\}) =$$

Min

$$\left\{ \begin{array}{l} K=2 : C(1,2) + \underline{g(2\{3,4\})} \\ K=3 : C(1,3) + \underline{g(3\{2,4\})} \\ K=4 : C(1,4) + \underline{g(4\{2,3\})} \end{array} \right\}$$



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1. $|S| = 0, \underline{S = \phi}$

$i = 2, 3, 4$

$g(2, \phi) = C(2, 1) = 5$

$g(3, \phi) = C(3, 1) = 6$

$g(4, \phi) = C(4, 1) = 8$

$V_0 = 1$

$J(2, \phi) = 1$

$J(3, \phi) = 1$

$J(4, \phi) = 1$

$2 \rightarrow 1$

$3 \rightarrow 1$

$4 \rightarrow 1$



Topic : Dynamic Programming :(DP)

2. $|S| = 1$

$$\begin{aligned} g(2, \{3\}) &= C(2, 3) + g(3, \phi) \\ &= 9 + 6 = \underline{15} \quad \checkmark \end{aligned}$$

$$g(2, \{4\}) = C(2, 4) + g(4, \phi) = 10 + 8 = 18 \quad \checkmark$$

$$g(3, \{2\}) = C(3, 2) + g(2, \phi) = 13 + 5 = \underline{18}$$

$$g(3, \{4\}) = C(3, 4) + g(4, \phi) = 12 + 8 = \underline{20}$$

$$J(2, \{3\}) = 3$$

$$J(2, \{4\}) = 4$$

$$J(3, \{2\}) = 2$$

$$J(3, \{4\}) = 4$$



Topic : Dynamic Programming :(DP)

$$g(4, \{2\}) = C(4, 2) + g(2, \phi)$$

$$= 8 + 5 = \underline{13}$$

$$g(4, \{3\}) = C(4, 3) + g(3, \phi)$$

$$= 9 + 6 = \underline{15}$$

$$|S| = 0$$

$$|S| = 1$$

$$|S| = 2$$

$$\underline{\underline{\quad\quad\quad}}$$

$$J(4, \{2\}) = 2$$

$$J(4, \{3\}) = 3$$



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3. $|S| = 2$

$$g(2, \{3, 4\}) = \min \begin{cases} K = 3: C(2, 3) + g(3, \{4\}) \\ K = 4: C(2, 4) + g(4, \{3\}) \end{cases}$$

$$= \min \begin{cases} K = 3: 9 + 20 \\ K = 4: 10 + 15 \end{cases} \quad \text{From prev. subproblem}$$

$$J(2, \{3, 4\}) = 4$$

$$= \min \binom{29}{25} = 25 \quad \checkmark$$



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$$g(3, \{2, 4\}) = \min \begin{cases} K=2: C(3, 2) + g(2, \{4\}) \\ K=4: C(3, 4) + g(4, \{2\}) \end{cases}$$

$$= \min \begin{cases} K=2: 13 + 18 \\ K=4: 12 + 13 \end{cases}$$

$$J(3, \{2, 4\}) = 4$$

$$= \min \begin{pmatrix} 31 \\ 25 \end{pmatrix} = 25$$



Topic : Dynamic Programming :(DP)

Q 31.

$$g(4, \{2, 3\}) = \min \begin{cases} K=2: C(4, 2) + g(2, \{3\}) \\ K=3: C(4, 3) + g(3, \{2\}) \end{cases}$$

$$= \min \begin{cases} K=2: 8 + 15 \\ K=3: 9 + 18 \end{cases}$$

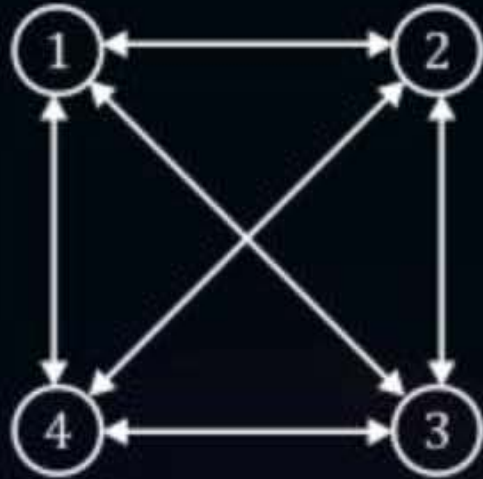
$$J(4, \{2, 3\}) = 2 \\ = \min \begin{cases} K=2: 23 \\ K=3: 27 \end{cases} \\ = 23$$

$$\underline{\underline{|S| = 3}}$$



Topic : Dynamic Programming :(DP)

E.g.



Cost adj. Matrix

C	1	2	3	4
1	0	10	15	20
2	5	0	9	10
3	6	13	0	12
4	8	8	9	0

Bottom - up

Optimal problem

$$g(1\{2,3,4\}) = \text{Min}$$

$$\left\{ \begin{array}{l} K=2 : C(1,2) + g(2\{3,4\}) \\ K=3 : C(1,3) + g(3\{2,4\}) \\ K=4 : C(1,4) + g(4\{2,3\}) \end{array} \right\}$$

$$|S|=3$$

$$J(1, \{2,3,4\}) = \underline{\underline{2}}$$

$$\downarrow$$
$$|S|=2$$



Topic : Dynamic Programming :(DP)

$$g(1, \{2, 3, 4\}) = \min \left\{ \begin{array}{l} K=2: C(1,2) + g(2, \{3, 4\}) \\ K=3: C(1,3) + g(3, \{2, 4\}) \\ K=4: C(1,4) + g(4, \{2, 4\}) \end{array} \right\}$$

$$= \min \left\{ \begin{array}{l} K=2: 10 + \underline{25} \\ K=3: 15 + \underline{25} \\ K=4: 20 + \underline{23} \end{array} \right\}$$

$$= \min \left\{ \begin{array}{l} K=2: \underline{35} \\ K=3: \underline{40} \\ K=4: \underline{43} \end{array} \right\} = \underline{35} \checkmark$$



Topic : Dynamic Programming :(DP)

#Q. How to determine the path/tour which we got min cost?

1 → 2 → 4 → 3 → 1

^{reported}
Tour ~~repeated~~ by TSP

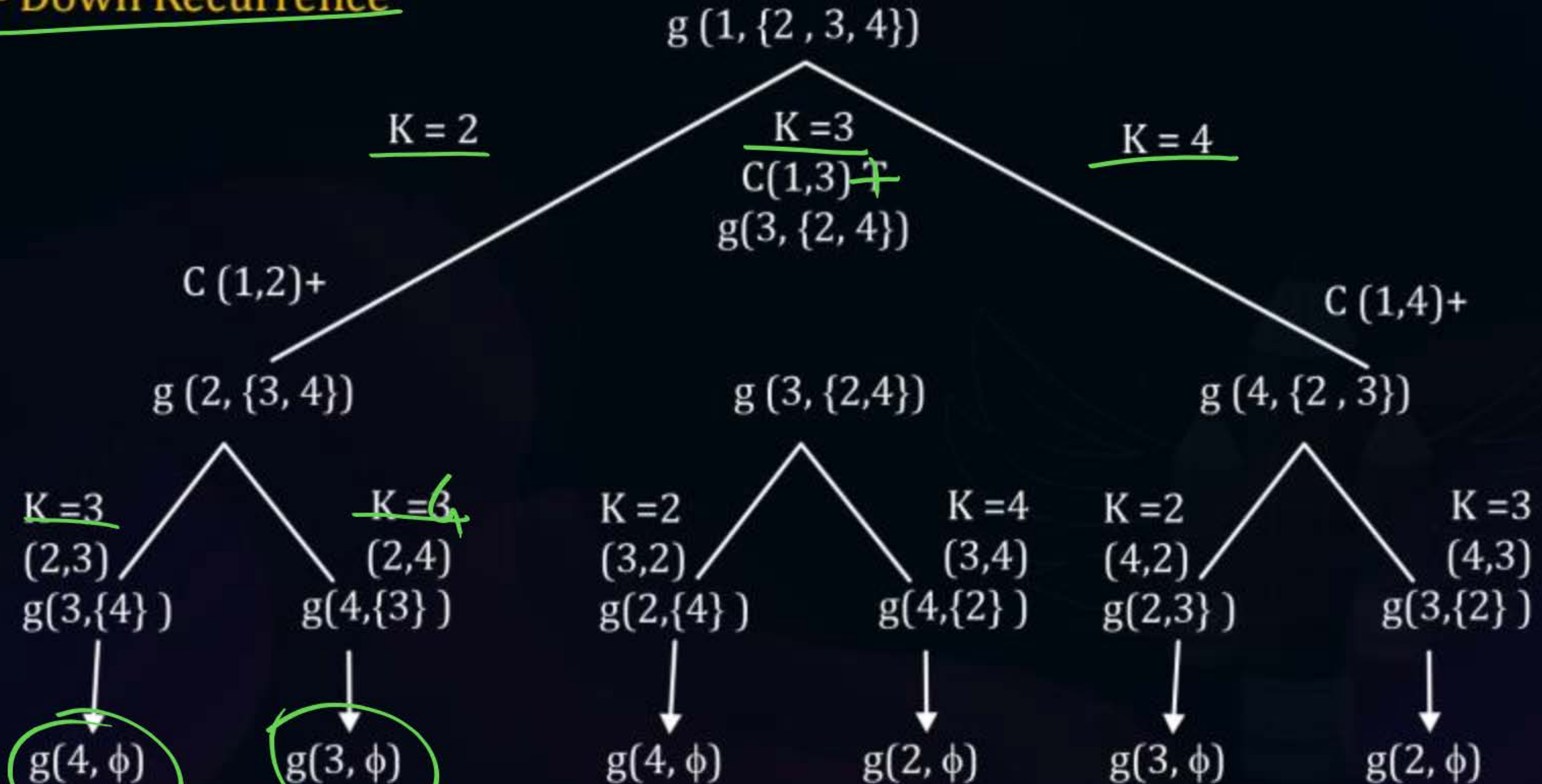
Cost = 35

[1 → 2 → 4 → 3 → 1]



Topic : Dynamic Programming :(DP)

Top- Down Recurrence





Topic : Dynamic Programming :(DP)

Held-Karp Algorithm :- DP based Algo to solve the travelling salesman problem (TSP)

1. Time complexity = $O(2^n * n^2)$
2. Space complexity $O(2^n * \sqrt{n})$

n = no. of cities./vertices

Note.. TSP is one of those problems for which there is ~~number polynomial~~ ^{no polynomial} TC (as of now)



Topic : Dynamic Programming :(DP)

P48

#Q. Dijkstra's SSSP Algorithm when run from vertex a in the below graph, computes the shortest path distance to?

HW

A

Only vertex a

B

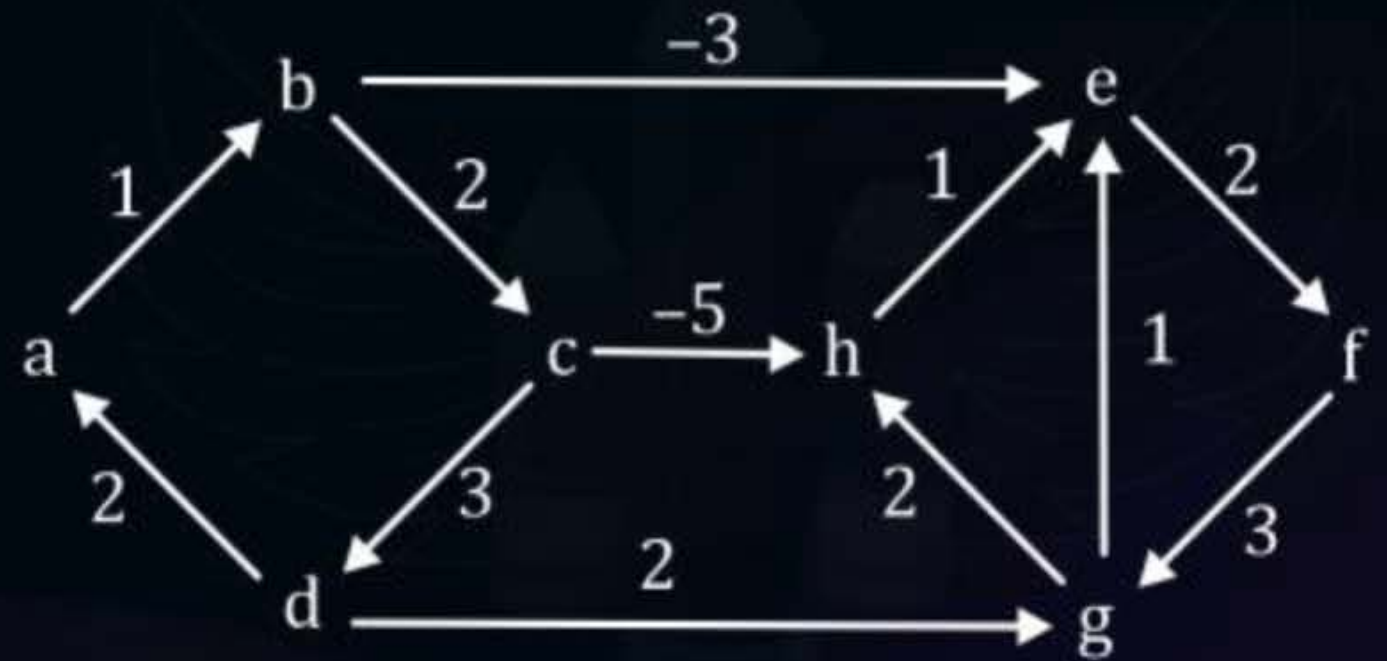
Only vertices a, e, f, g, G

C

Only vertices a, b, c, d

D

All the vertices.





Topic : Dynamic Programming :(DP)

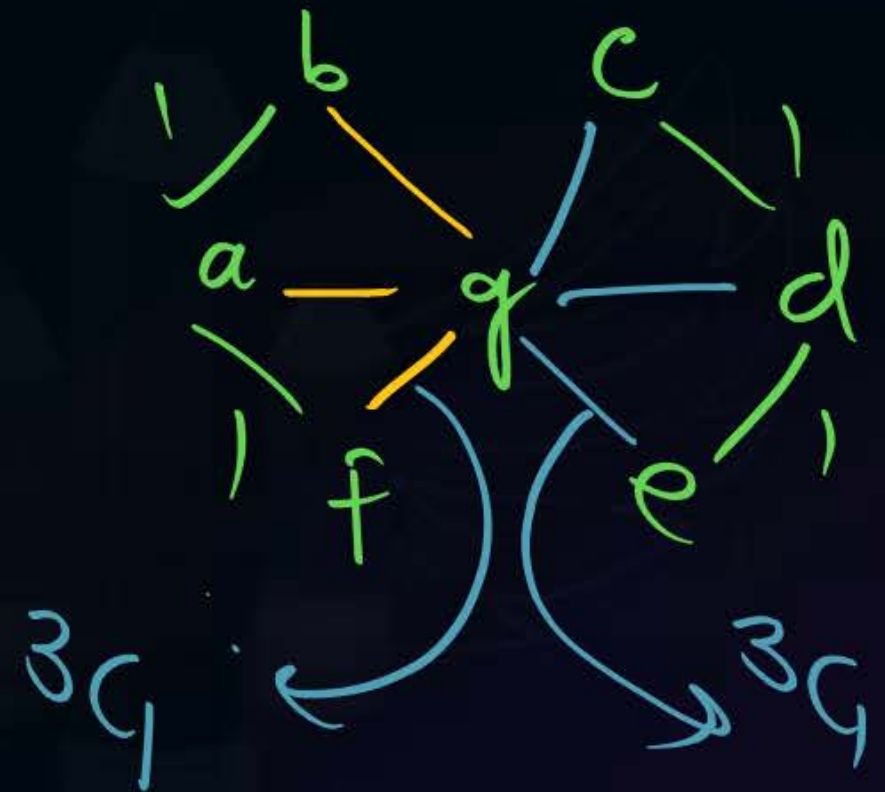
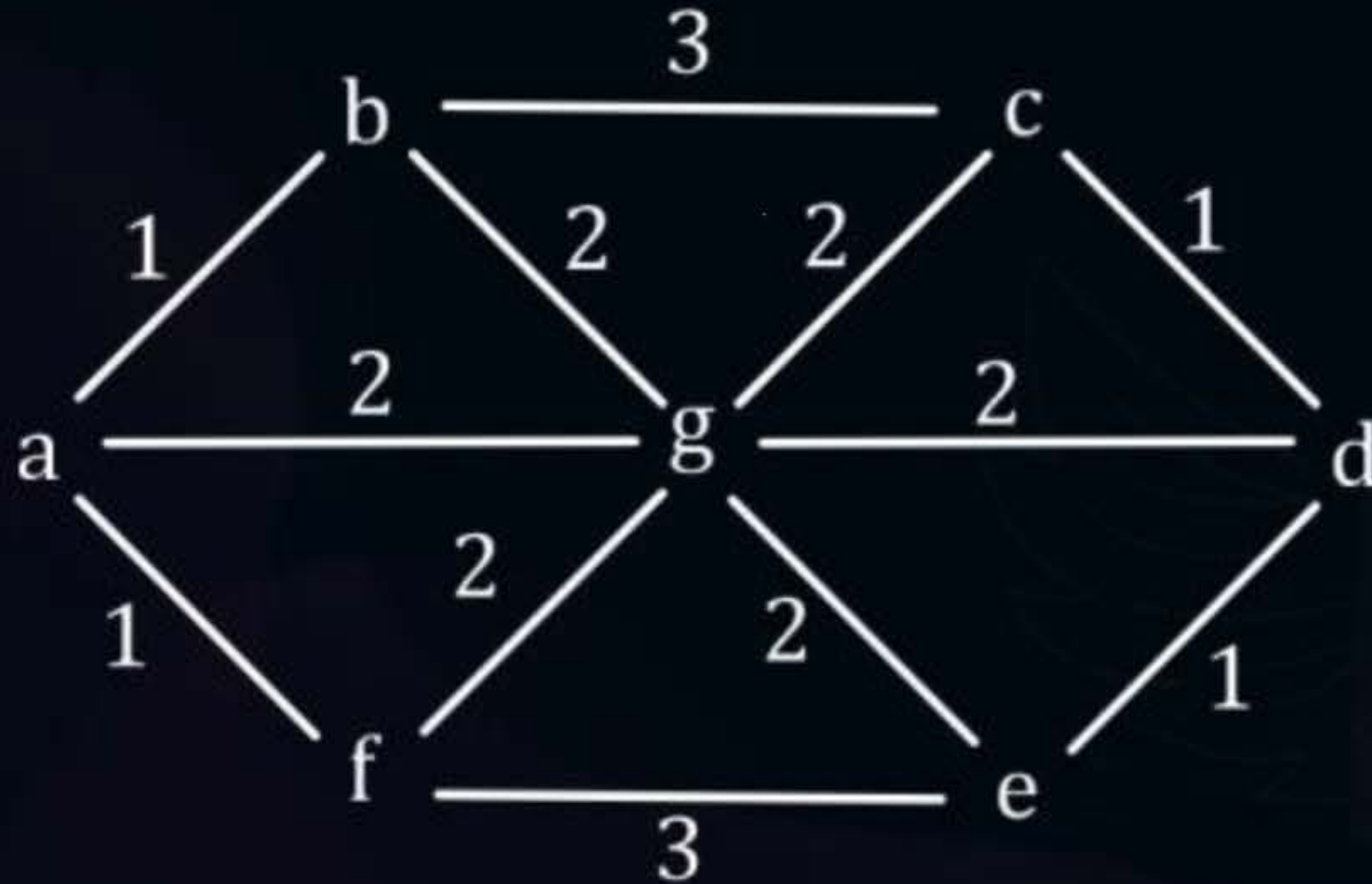
#Q. How many distinct MCSTs?

10%

Gate 2024

2M

$$\begin{aligned}\text{Total} &: 3C_1 * 3C_1 \\ &= 3 \times 3 \\ &= \textcircled{9}\end{aligned}$$





2 mins Summary



Topic

Topic

Topic

Topic

Topic

TSP

PYQs



THANK - YOU